Impact of a Pharmaceutical Care Program in a Community Pharmacy on Patients with Dyslipidemia

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BACKGROUND: Inappropriate use of medications is a significant problem in health care today. A possible solution to this problem may be achieved through better control of patients' drug therapy.

OBJECTIVE: To design a pharmaceutical care program for dyslipidemic patients within a community pharmacy setting that provides education in the areas of medication compliance and lifestyle modifications, while emphasizing the importance of achieving cholesterol goals to ensure improvement in quality of life.

METHODS: Patients at an outpatient pharmacy volunteered to be surveyed for 16 weeks. Although both the intervention and control groups were surveyed, the randomly selected intervention group was interviewed more frequently and more comprehensively. Cholesterol, triglycerides, glucose, weight, risk factors, drug-related problems (DRPs), and quality of life were measured via a survey at the onset of the study and continually measured until the study's conclusion.

RESULTS: In the intervention group, 26 DRPs were detected, of which 24 were resolved; in the control group, 26 DRPs were detected, of which 5 were resolved. When comparing initial and final blood cholesterol levels in the intervention group, the mean decrease was 27.0 ± 41.1 mg/dL ($p = 0.0266$); in the control group, the average blood cholesterol level decreased by a mean of 1.4 ± 37.2 mg/dL ($p = 0.6624$). In the intervention group, the triglyceride level decreased an average of 50.5 ± 80.3 mg/dL ($p = 0.0169$), while the control group experienced a mean triglyceride level increase of 29.6 ± 118.5 mg/dL ($p = 0.1435$). As a result of the intervention, the quality of life in the intervention group was improved.

CONCLUSIONS: Short-term pharmaceutical care plans developed in a retail pharmacy within the proper setting may contribute to improved blood lipid values, cardiovascular disease risk factors, and patients' quality of life.

KEY WORDS: community pharmacy, drug-related problems, dyslipidemia, pharmaceutical care.

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Inappropriate use of medications is a significant problem in health care today. A possible solution is better control of patients' drug therapy, which may also provide the perfect opportunity for pharmacists to become more proactive in their care of patients. When the role of a community pharmacy is broadened to include the full realm of pharmaceutical care, patients can achieve and improve their quality of life and longevity, and the potential exists to restore pharmacy to one of the most trusted professions.[1]

Pharmaceutical care is a practice in which the pharmacist takes full responsibility and accountability for a patient's drug-related needs. When pharmacists provide pharmaceutical care, they use all their knowledge and skills to the benefit of the patients while also providing an advanced level of care over an extended period of time.[2,3] The main goal of pharmaceutical care is to meet a social need. For example, patients need assistance in obtaining maximum benefits from the medications they use through the identification, solution, and prevention of drug-related problems (DRPs).[4]

A DRP is an undesirable event experienced by a patient that is involved or suspected to be involved in drug therapy, which may or may not interfere with the patient's desired outcome.[3] The clinical goal for the prevention and solution to a DRP is to avoid drug-related morbidity and mortality as well as its economic consequences.[1]

Chile has experienced a demographic transition characterized by increased life expectancy, which has led to a larger aging population. As a result, non-transmissible chronic diseases constitute a new challenge for the country's public health system.[4] The main nutritional problems

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appear to be obesity and dyslipidemia, which are risk factors for cardiovascular disease. In Chile, cardiovascular diseases are the country's leading cause of death, accounting for 27.1% of overall mortality.[5,6]

Cardiovascular disease risk factors, such as smoking, elevated serum cholesterol, obesity, and lack of physical activity, are considered modifiable through medical therapy or changes in behavior. However, elevated cholesterol is also recognized as a major modifiable risk factor for the development of cardiovascular disease.[7]

The population in Chile has shown a progressive increase in cholesterol plasma levels.[6,8] Dyslipidemia is defined as a series of metabolic disturbances reflected by abnormal blood concentrations of cholesterol and/or triglycerides and is associated with health risk factors. The importance of dyslipidemias lies in their causal relationship to atherosclerosis, especially manifested in ischemic heart disease.

The treatment of dyslipidemias is based mainly upon patients' lifestyle modifications such as changes in eating habits, increase in physical activities, and decrease or cessation of other risk factors such as alcohol intake, smoking habits, and excess weight.[9] Drug therapy is indicated for dyslipidemia when the therapeutic response to diet is insufficient or when lifestyle changes do not effectively reduce cholesterol.

The cause-and-effect relationship between blood cholesterol levels and the relative risk for developing coronary heart disease was demonstrated in the 1980s.[10] More recent studies have shown that reduction in low-density lipoprotein cholesterol levels reduces coronary artery disease events and total mortality.[11,12] The data on these cause-and-effect relationships support creating a pharmaceutical care program for dyslipidemic patients that would identify and solve DRPs and provide education about the disease and healthy lifestyles in order to support drug therapy, with the final aim being to improve patients' quality of life.

Methods

Chile has close to 1400 pharmacies, serving a total population of approximately 15 million people. This study was completed in a community pharmacy in Santiago, Chile. The pharmacy has an established counseling area designed especially for personalized counseling and basic physical assessment. The study was conducted by a pharmacist who was dedicated solely to this study and who was trained specifically for the purposes of this study. The sample selection was made up of the pharmacy's customers. In the recruitment phase, the first contact with the patients took place when customers purchased their medication or requested blood cholesterol or triglyceride analysis (services provided by the pharmacy).

The study was explained to the patients who were interested in participating. Institutional review board approval was not sought, although the pharmacist did obtain informed consent from each participant upon enrollment in the study. The inclusion criteria for the sample selection were minimum age 18 years, both genders, clinical diagnosis of dyslipidemia, and patient currently being treated for this disorder. The exclusion criteria were patients with communication difficulties, pregnancy, and self-medicating patients. The patients admitted to the trial were randomly divided into a control group and an intervention group. The control group received normal counseling, while the intervention group received a comprehensive pharmaceutical plan and scheduled follow-up.

Both groups were surveyed for 16 weeks. During this time period, the control group had 2 interviews, while the intervention group was interviewed 5 times. Patients received a card with the dates of the interviews; the period of time between the first and last interview was the same for both groups. Each interview had an assigned time of 20–25 minutes.

Pharmacist intervention included obtaining total blood cholesterol and triglyceride levels as well as teaching patients about the role of cholesterol in illness and health, explaining risk factors associated with cardiovascular disease, and providing education/counseling regarding medication. Each interview was used to determine, resolve, or prevent DRPs based on the information obtained from the patient in addition to their medical records. When the pharmacist detected a DRP, the patient was referred to a physician with the purpose of solving the problem.

To determine drug adherence, a visual analog scale was used. The patients made a self-assessment using a scale scored between 0 and 10, with zero indicating total nonadherence and 10 indicating that the patient comprehensively followed the medical instructions.

In each interview, manual records for the intervention and control groups were completed. These records included personal data such as drug therapy, adherence to treatment, lifestyles, DRPs, and the following clinical variables: total blood cholesterol level, triglyceride level, and body mass index (BMI). Those values were measured at the first, third, and fifth interview in the intervention group and in the 2 corresponding interviews in the control group. An Accutrend GCT (Roche Diagnostics Laboratories, Mannheim, Germany) device was adopted for this study, using a capillary blood drop obtained from the fasting patient's finger. Patients' weights were recorded at the beginning and end of the program using the same calibrated digital scale. With weight and height data, the patient's BMI was calculated to classify obesity levels.[13]

A solved DRP was defined as an achieved health goal and the correction or resolution of suspected drug-related morbidity. An unresolved DRP was one that did not accomplish the proposed health goals, with persistence of the abnormal situation that started the suspected DRP. DRPs identified in this study were categorized according to the classification established by Cipolle et al.[2] (Table 1).

The Health Short-Form–36 survey was used to determine the patients' quality of life.[14] At the first and final interviews at the pharmacy, the survey was given to patients in both groups to be answered at home and returned at the next visit to the pharmacy. The survey evaluated the patient's functional status, emotional well-being, and general health status based on the answers to 11 questions.

Brochures specifically designed for this program were based at a reading and language level suitable for our patient demographic, with the purpose of educating patients about their disease as well as healthy lifestyle options (eg, balanced diet, practical advice to lower cholesterol). These brochures were given and explained to only the patients of the intervention group in the third and fourth interview, respectively, and followed up with a question and answer session to ensure full understanding of items discussed.

At the end of the program, patients in the intervention group were given a 10-question evaluation sheet to answer at home regarding their level of satisfaction with the program.

STATISTICS

Descriptive statistics of the sample group were applied to classify both groups. The Shapiro–Wilks test was applied to determine the following variables: total blood cholesterol, triglycerides, weight, and BMI. According to the results, parametric (Student's *t*-test for paired data) and nonparametric (Wilcoxon signed ranks test for non-paired data, signed test for paired data) tests were applied. Statistical analysis was performed with the STATA version 7.0 program.

The quality-of-life variable was evaluated through a quantitative analysis using the SPSS version 8.0 program. This analysis was applied to each patient at the beginning and end of the program. Statistical significance was set at $p < 0.05$.

Results

Forty-two volunteers completed this program. The groups were randomly assigned, with 23 patients in the intervention group and 19 in the control group. The mean ± SD age for men and women was 64 ± 10 and 66 ± 11 years, respectively. Gender distribution for the total sample size was 81% female, 19% male. The education level of the patients in the study revealed that 33.3% graduated from a university, 16.7% graduated from technical institutes, 23.8% had completed high school, 21.4% had completed some high school, and 4.8% did not complete elementary school. With respect to concomitant diseases, 52% of the patients had been diagnosed with arterial hypertension.

The most commonly used pharmacologic treatment for dyslipidemia in both groups was the hydroxymethylglutaryl coenzyme A (HMG-CoA) reductase inhibitors (76.9% and 94.7% for the intervention and control groups, respectively). Atorvastatin was the most commonly used drug for dyslipidemia therapy in both groups.

In general, 72.8% of the intervention group experienced a decrease in blood cholesterol levels, 22.7% experienced an increase, and 4.5% experienced no change. In the control group, cholesterol decreased in 33.3% of patients, with an increase in 61.1% and with 5.6% showing no change.

Cholesterol levels in the intervention group decreased by a mean of 27.1 ± 41.1 mg/dL ($p = 0.0266$), from 205.1 ± 44.7 to 178.1 ± 31.1 mg/dL. In the control group, cholesterol levels decreased by a mean of 1.4 ± 37.2 mg/dL, from 203.2 ± 40.6 to 199.1 ± 37.6, which was not statistically significant ($p = 0.6624$). No patient had a cholesterol level >300 mg/dL at any point during the study based on the limit detection value of the device used in this study.

In the intervention group, 77.3% of the patients lowered their blood triglyceride level between the initial and final measurements, while 22.7% experienced an increase. The triglyceride level for this group decreased by an average of 50.5 ± 80.3 mg/dL ($p = 0.0169$), from 190.7 ± 88.7 to 140.3 ± 47.6 mg/dL. In the control group, 27.8% lowered their triglyceride level, 66.7% increased it, and 5.5% showed no changes. There was a mean triglyceride increase for this group of 29.6 ± 118.5 mg/dL, from 163.6 ± 116.4 to 193.2 ± 108.0 mg/dL, which was not statistically significant ($p = 0.1435$). No patient had a triglyceride level >600 mg/dL, which is the detection limit of the device.

At the end of the program, body weight within the intervention group decreased an average of 1 ± 1.3 kg, while in the control group, the average weight increased by 1.1 ± 2.6 kg. There was no significant difference between the initial BMI measurement within the 2 groups ($p = 0.2384$), but there was a significant decrease of 0.4 ± 0.5 kg/m² within the intervention group ($p = 0.0176$).

Only 1 of the 22 patients within the intervention group had a mild smoking habit (3 cigarettes/day), which persisted until the end of the program. Six of the control group patients smoked at the begining of the program; at the end of the program, 4 had increased the number of cigarettes smoked per day.

Fifty percent of the intervention group and 72% of the control group did not drink alcoholic beverages. These numbers persisted until the end of the program, although 2 patients in the intervention group stopped drinking alcohol during the study period and 1 patient decreased his alcohol intake.

Approximately 86.4% of the intervention group and 83.4% of the control group were sedentary at the beginning of the program. We considered a person to be physically active if they performed exercise for at least 30 minutes 3 times weekly; volunteers falling below that amount were considered sedentary. At the end of the survey, only 45% of the intervention group were sedentary, while the number in the control group remained unchanged.

The intervention group had 26 DRPs in 12 patients, of which 24 (92.3%) were solved, while the control group had 26 DRPs in 14 patients, of which 5 (19.2%) were solved without intervention. With respect to the distribution of the 52 DRPs detected in all patients, the most frequent proved to be DRP 4 (38.5%), followed by DRP 2 (26.9%; Table 1).

The quality-of-life additive index was similar in both groups at the beginning of the program, with no statistical difference between them ($p = 0.57$). However, at the end of the study, there was a significant difference in the quality-of-life index between the 2 groups ($p = 0.002$).

The program was well received by the patients, as expressed by their degree of satisfaction. On a scale of 1–7, 77.8% of the patients rated their level of satisfaction with the program at the maximum (7.0), with the rest choosing 6.0.

Table 1. Classification of Drug-Related Problems[2]

DRP 1	The patient has a medical condition that requires the initiation of new or additional drug therapy.
DRP 2	The patient is taking drug therapy that is unnecessary given his or her present condition.
DRP 3	The patient has a medical condition for which the wrong drug is being taken.
DRP 4	The patient has a medical condition for which too little of the correct drug is being taken.
DRP 5	The patient has a medical condition resulting from an adverse drug reaction.
DRP 6	The patient has a medical condition for which too much of the correct drug is being taken.
DRP 7	The patient has a medical condition resulting from not taking the drug appropriately.

DRP = drug-related problem.

Discussion

Fifty-two percent of the patients who entered the study for dyslipidemia also had hypertension, which confirms data pointing to the progressive increase in hypertension over time, reaching >50% of people older than 65 years. The gender distribution for the total sample size is indicative of the pharmacy's clientele and also shows that women were more motivated to participate in this type of study.

The study also revealed that HMG-CoA reductase inhibitors were the most frequently used class of dyslipidemia medication. This is consistent with current literature that classifies this drug class as the most frequently prescribed class of medication for the treatment of dyslipidemia.[15]

In the intervention group, 72.8% were able to lower their cholesterol level to better category ranges, while 22.7% experienced an increase in cholesterol levels. This indicates that the program fulfilled the goal of lowering plasma lipids of patients whose levels were above normal. Additionally, all of the patients in the intervention group with triglyceride levels >200 mg/dL were able to lower this level and improve the classification category.

The continuous recording of patients' medication use, drug therapy changes, and measurements of dyslipidemia-related clinical parameters allowed the identification of existing or potential DRPs and, thus, facilitated the introduction of solution-oriented actions. The most frequent DRP in the patients studied was related to subtherapeutic drug dosage, usually secondary to an inadequate frequency of administration. High drug costs and failure to remember to take the drug due to the intake of several other concomitant drugs were the most frequent causes of nonadherence to drug therapy, especially with dyslipidemia medications. Over 50% of the patients in each group experienced DRPs. These results highlight the need to establish programs such as this in an effort to meet the social and economic need of the public and, in particular, the elderly.

Only a few studies have demonstrated positive outcomes from pharmaceutical care in an ambulatory care community setting.[16-18] Our study demonstrates 2 main points: first, it shows that it is possible for patients to receive advanced pharmaceutical care in a community pharmacy, leading to improved health outcomes, improved quality of life, and patient satisfaction; and second, the study points out the importance of the pharmacist in improving health care, regardless of the clinical setting. The pharmacist's role as an integral member of the healthcare team has once again been validated with this study, also demonstrating that it is possible to give customers pharmaceutical care in a community pharmacy. The pharmacist can play an important role in improving the satisfaction and quality of life of patients with hyperlipidemia.[19-21]

Not only were significant laboratory values improved, resulting in the possibility of better patient outcomes, but patients also expressed a high level of satisfaction with their health care. These 2 key components demonstrate a justification for programs such as this one to be adopted within pharmacies.

Summary

This study showed that the provision of pharmaceutical care contributed to the improvement of blood lipid values and quality of life in patients with dyslipidemia. The study also demonstrated the important role of the pharmacist in the identification and resolution of DRPs through optimized treatment regimens. The patients' interest in this type of service was demonstrated by the number of volunteers who participated in the study, as well as the favorable results within the satisfaction survey. This study offers clear evidence that the community pharmacy has a strategic role within the healthcare system as a means to establish pharmaceutical care in Chile.

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EXTRACTO

ANTECEDENTES: El uso inapropiado de medicamentos es un problema significativo en la salud de hoy en día. Una posible solución a este problema se puede lograr a través de un mejor control de la terapia con medicamentos.

OBJETIVO: Diseñar un programa de atención farmacéutica para pacientes con dislipidemia en farmacia comunitaria que entregue educación en áreas relacionadas con la adherencia al tratamiento y modificaciones del estilo de vida, haciendo énfasis en la importancia de alcanzar concentraciones adecuadas de colesterol y asegurar así un mejoramiento en la calidad de vida.

MÉTODOS: Los pacientes fueron voluntarios y clientes de una farmacia comunitaria los cuales fueron estudiados durante 16 semanas. Aunque ambos grupos, el grupo control y el grupo intervenido, fueron estudiados, el grupo intervenido seleccionado al azar fue entrevistado más frecuentemente y de manera más completa. Se midió el colesterol, triglicéridos, glucosa, peso, factores de riesgo, problemas relacionados con medicamentos (PRM), y la calidad de vida a través de entrevistas realizadas al comienzo del estudio y que se mantuvieron hasta el final del mismo.

RESULTADOS: En el grupo intervenido, se detectaron 27 PRM de los cuales 24 fueron resueltos. Por su parte, en el grupo control, se detectaron 26 PRM de los cuales se resolvieron 5. Al comparar las concentraciones de colesterol sanguíneo inicial y final, se observó una disminución significativa en el grupo intervenido de 27.0 ± 41.1 mg/dL ($p = 0.0266$), mientras que en el grupo control, estas concentraciones disminuyeron en promedio 1.4 ± 37.2 mg/dL ($p = 0.6624$). En el grupo intervenido, las concentraciones de triglicéridos disminuyeron en promedio 50.5 ± 80.3 mg/dL ($p = 0.0169$), mientras que en el grupo control, se observó un aumento promedio de 29.6 ± 118.5 mg/dL ($p = 0.1435$). Como resultado de la aplicación del programa , la calidad de vida mejoró significativamente en el grupo intervenido.

CONCLUSIONES: Este estudio demostró que programas de atención farmacéutica desarrollados en farmacia comunitaria pueden contribuir a mejorar los valores de lípidos en la sangre, los factores de riesgo cardiovascular, y la calidad de vida de los pacientes con dislipidemia.

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RÉSUMÉ

INTRODUCTION: L'utilisation inappropriée de médicaments est considérée comme un problème majeur dans les soins de santé. Une solution possible à ce problème peut consister en un meilleur contrôle de la thérapie d'un patient.

OBJECTIF: Développer un programme de soins pharmaceutiques pour des patients dyslipidémiques au sein d'une pharmacie communautaire. Ce programme impliquerait une composante éducationnelle pour améliorer l'adhérence au traitement et pour favoriser les modifications des habitudes de vie tout en mettant l'emphase sur le contrôle du cholestérol en vue d'améliorer la qualité de vie.

MÉTHODOLOGIE: Les patients d'une pharmacie communautaire ont consenti à une participation durant une période de 16 semaines. Bien que les 2 groupes de patients, ceux ayant subi l'intervention et ceux du groupe contrôle, aient été interrogés, ceux du groupe intervention l'ont été plus souvent et plus rigoureusement. Les taux de cholestérol et de triglycérides, la glycémie, le poids, les facteurs de risque, et les problèmes reliés à la pharmacothérapie (PRP) et la qualité de vie furent évalués par un sondage en début d'étude et mesuré continuellement jusqu'à la fin de celle-ci.

RÉSULTATS: Dans le groupe intervention, 27 PRP furent détectés, desquels 24 furent résolus, alors que dans le groupe contrôle, 26 PRP furent détectés et seuls 5 furent résolus. Lorsqu'on compare le taux de cholestérol initial et final dans le groupe intervention, une réduction moyenne de 27 ± 41.1 mg/dL fut observée ($p = 0.0266$) alors que dans le groupe contrôle, la réduction moyenne du taux de cholestérol a atteint 1.4 ± 37.2 mg/dL ($p = 0.6624$). Le taux de triglycérides du groupe intervention fut réduit de 50.5 ± 80.3 mg/dL ($p = 0.0169$), alors que celui du groupe contrôle a augmenté de 29.6 ± 118.5 mg/dL ($p = 0.1435$). Considérant ces résultats, la qualité de vie des patients du groupe intervention s'est améliorée.

CONCLUSIONS: Cette étude démontre que l'élaboration de plans de soins à court terme dans une pharmacie communautaire peut contribuer à améliorer le bilan lipidique, les facteurs de risque cardiovasculaire, et la qualité de la vie des patients.

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